ML Regression Project: Life Expectancy Prediction

Dataset: Life Expectancy Data

1. Introduction

Healthcare and population health analytics play a pivotal role in understanding the overall well-being and development of nations. Among various public health indicators, **life expectancy** is one of the most significant, as it reflects a country's socioeconomic conditions, quality of healthcare systems, educational attainment, and overall disease burden.

1. Objective

The objective of this project is to build a Machine Learning regression model to predict Life
Expectancy using health-related indicators such as Adult Mortality, Infant Deaths, BMI, and GDP.

2. Dataset Description

The dataset used is 'Life Expectancy Data.csv'. It contains country-level health, economic, and
demographic indicators collected over multiple years.

3. Features and Target

Type
Columns
Features (X)
Adult Mortality, infant deaths, BMI, GDP
Target (y)
Life Expectancy

4. Machine Learning Model

A Multiple Linear Regression model was used. The dataset was split into training and testing sets
(80% training and 20% testing). The model learns the relationship between the selected features
and life expectancy.

5. Model Evaluation

The model performance was evaluated using R^2 Score and Mean Absolute Error (MAE). The
achieved R^2 score of approximately 0.64 indicates that the model explains around 64% of the
variance in life expectancy. The MAE of about 4 years shows reasonable prediction accuracy for
real-world health data.

6. Coding

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error

# ==========================
# Load Dataset
# ==========================
df = pd.read_csv("Life Expectancy Data.csv")

print("Dataset loaded successfully ✅")

# ==========================
# Clean column names
# ==========================
df.columns = df.columns.str.strip()

print("\nCleaned Columns:\n", df.columns)

# ==========================
# Required Columns
# ==========================
required_columns = [
    'Life expectancy',
    'Adult Mortality',
    'infant deaths',
    'BMI',
    'GDP'
]

df = df[required_columns]
df = df.dropna()
```

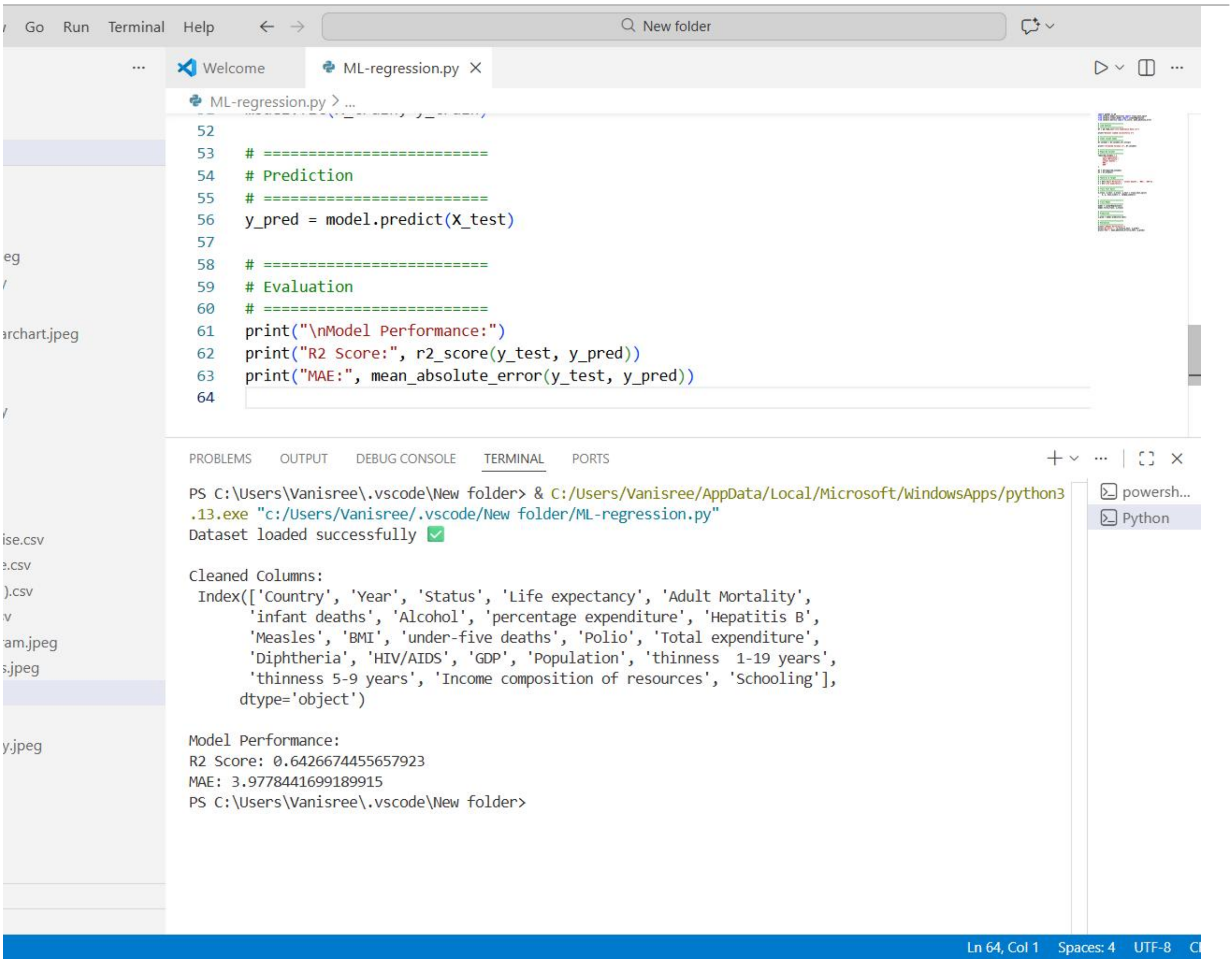
```
# ===========================
# Features & Target
# ===========================
X = df[['Adult Mortality', 'infant deaths', 'BMI', 'GDP']]
y = df['Life expectancy']

# ===========================
# Train Test Split
# ===========================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ===========================
# Train Model
# ===========================
model = LinearRegression()
model.fit(X_train, y_train)

# ===========================
# Prediction
# ===========================
y_pred = model.predict(X_test)

# ===========================
# Evaluation
# ===========================
print("\nModel Performance:")
print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
```



7.Conclusion

This project demonstrates the application of machine learning regression techniques to real-world
health data. The model provides meaningful insights into how mortality, economic conditions, and
health indicators affect life expectancy.